Certified Local Transferability for Evaluating Adversarial Attacks

Minyu Chen, Jingyang Li, Ling-I Wu, and Guoqiang Li, *Member, IEEE*

***Abstract*—Deep neural networks (DNNs) are known to be vulnerable to adversarial examples. Though adversarial attacks show effectiveness in misleading models, most attack methods are designed to poison a specific image. To investigate the actual effect on the feature space, we introduce the concept of the certified local transferable region. This is a connected area of inputs where we can mathematically guarantee that a single adversarial perturbation will successfully fool the model. The size of this region is a metric to evaluate the local transferability of perturbations. We present a novel method, reverse attack oracle-based search (RAOS), to estimate the maximum size of this region. Our approach efficiently searches for the largest possible vulnerable area around an original input by iteratively refining its boundaries. Each step is guided with a minimal distance attack algorithm and proven with state-of-the-art verifiers. We conduct empirical experiments to evaluate various attacks on different model structures and adversarial training scenarios. We show the advantage of our proposed metric over existing ones and demonstrate its utility in exploring the robustness of neural networks.**

***Impact Statement*—While DNNs have demonstrated remarkable success across real-world applications, their vulnerability to adversarial attacks remains a critical limitation, particularly in safety-sensitive domains such as autonomous driving and medical diagnosis. Existing adversarial attack methods primarily focus on perturbing discrete input points, but their effectiveness in continuous input spaces remains poorly understood. Our research bridges this gap by introducing a novel framework that quantifies the susceptibility of entire local neighborhoods to adversarial perturbations. By identifying a certified maximum vulnerable region around each input, we provide a more comprehensive measure of local attack transferability. This approach enhances our understanding of adversarial robustness, demonstrating that current defenses relying on input transformations may be insufficient in countering attacks that impact broader input regions. Our findings pave the way for stronger, region-based defense mechanisms, ensuring the deployment of more resilient artificial intelligence models in high-stakes environments.**

***Index Terms*—Adversarial attacks, certified local transferability (CLT), deep neural networks (DNNs), neural network robustness.**

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The authors are with the School of Computer Science, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: minkow@sjtu.edu.cn; lijjjjj@sjtu.edu.cn; wulianyi1998@sjtu.edu.cn; li.g@sjtu.edu.cn).

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I. Introduction

SIGNIFICANT strides have been taken in deep neural networks (DNNs) for deployment across a spectrum of real-world systems. Regrettably, DNNs are vulnerable to potentially malicious adversarial examples, a susceptibility that curtails their integration in safety-critical applications such as autonomous driving [14] and medical diagnosis [15]. These adversarial examples, while typically imperceptible to humans, perturb neural networks into producing erroneous classification outcomes.

Under the white-box attack setting, where the attackers have access to the architecture and parameters of the target model, existing attack methods have demonstrated remarkable efficacy in terms of attack success rates (ASRs) [4], [17], [29], [31], even when pitted against adversarially trained models [17], [37], [42]. However, the established attack techniques primarily concentrate on generating adversarial examples for individual concrete input data points in discrete input space such as RGB values for image processing, and tokens for natural language processing. In deep learning applications with continuous input spaces, such as autonomous driving [16], creating frame-level, real-time adversarial examples with low delay is challenging. Attacks must have the capability to transfer across inputs that can be subtly altered over time. Furthermore, attackers may not always obtain accurate input; instead, noise can be introduced during the data collection phase of an attack. Nevertheless, from the perspective of defenders, the input transformation method [19], [32] is a representative defensive technique to reduce the ASR. By adding noises to the examples after the adversarial attack step and before the prediction step, adversarial attacks can also be fooled. Fig. 1 illustrates the lapse of attack methods in the proximal region of the original victim data point. Given a perturbation δ for shifting the victim data point x over the classification $f(x) = 0$, only inputs from a limited region can be affected by the perturbation δ.

These issues reveal a clear gap: attacks are typically designed for single, isolated points, but in the real world, an attacker's target is often a continuous range of similar inputs.

Therefore, instead of just testing a perturbation on a single input, we focus on a broader question: how vulnerable is the entire local area around that input to the same attack? Considering that the vulnerability to a single perturbation is kept within a relatively small region, we focus on searching a maximum region that is *certified* to be always vulnerable, to model the local transferability of any single perturbation,

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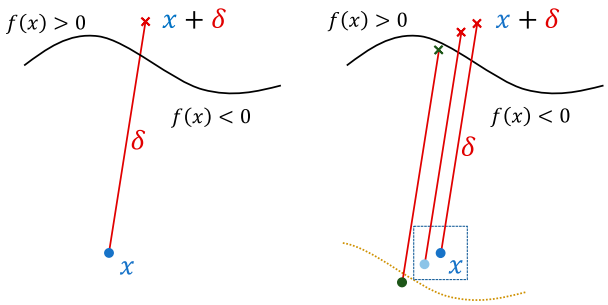


Fig. 1. Illustration of local transferability for adversarial attacks. The dark blue dot represents the original victim data point $\boldsymbol{x}$, the red lines stand for the introduced adversarial perturbations δ given to $\boldsymbol{x}$, and the red dot is the corresponding adversarial example. The black curve serves as the classification boundary. The perturbation can successfully attack any point (in light blue) in a proximal region (dotted blue line). However, for points out of this region such as the dot in green, the perturbation cannot always impact the targets.

instead of sampling proximal data points within the region to get a local ASR, which remains tolerant to attack failure.

To this end, we introduce the concept of *certified local transferability* (CLT) as a quantitative measure for analyzing the local transferability of adversarial attacks. This metric quantifies the maximum attainable volume of a connected vulnerable region in close proximity to the original attacked data point. Within this context, the term *vulnerable* refers to the property that *all* points within the region remain susceptible to misclassification by the target model when subjected to the same perturbation generated based on the original attacked data point. In practice, we investigate the transferability of attack methods in the ℓ_p spheres centered with the victim data points. Those attacks that can simultaneously impact a larger ℓ_p sphere around the original victim data point can be defined as attacks with better CLT. As the radius of ℓ_p spheres can denote their volume, the CLT metric can be simplified to the radii of the maximum vulnerable ℓ_p sphere.

To compute the CLT metric precisely, we first leverage neural network verifiers to *certify* the vulnerability of any ℓ_p sphere with a given radius. To avoid the uncertainty AND precision loss introduced by directly solving the optimization problem, we can transform the optimization problem into a search one by judging if ℓ_p spheres with different radii are vulnerable with a more precise verifier. Subsequently, we introduce a novel algorithm, called reverse attack oracle-based search (RAOS) to efficiently determine the possible maximum radius ϵ of vulnerable ℓ_p sphere by iteratively refining it with neural network verifiers. Specifically, during each iteration, we shrink the upper bound of ϵ by tracing back the attack steps of a minimal ℓ_p distance-based attack method. Simultaneously, we access a lower bound from state-of-the-art verifiers according to the maximum radius of the certified vulnerable ℓ_p sphere.

In this study, we explore the behavior of various typical attack methods on different model architectures under diverse training strategies via our proposed CLT. The proposed metric enables us to more precisely analyze the effect of attacks in the continuous space, which is worth discussing, especially in scenarios with continuous input space. Findings from experimental results demonstrate the value of our novel perspective on adversarial attacks and provide a better understanding of DNN robustness.

We outline our primary contributions as follows.

1) We propose the innovative concept of *CLT* to characterize a connected proximal region that is guaranteed to be vulnerable to the same perturbation. For the first time, this metric quantifies the impact that perturbations exert on the local regions surrounding the victim data points with the combination of neural network verification technologies.
2) We develop the RAOS, a novel and effective bound search framework, to estimate the maximum attainable volume of the connected vulnerable region. The algorithm shows its flexibility by allowing seamless integration with diverse neural network verifiers and supporting various additive perturbations across any ℓ_p-norm constraints, including widely utilized norms such as ℓ_1, ℓ_2, and ℓ_∞.
3) We conduct a thorough evaluation of typical attack methods using our novel metric, *CLT*. Our objective is twofold: to expand the scope of evaluating adversarial attacks beyond the sole consideration of ASRs and to offer a more nuanced perspective for evaluating these methods based on preimage analysis.

II. Related Work

In this section, we commence by providing a concise overview of classical adversarial attacks. Subsequently, we delve into the task of neural network certification. Within our exposition of the related work, we highlight the differentiating factors between our proposed notion of CLT and the existing body of research.

A. Adversarial Attacks

Gradient-based attacks utilize the gradients of the neural network's loss derived from the input data to generate adversarial perturbations. One prominent method is the fast gradient sign method (FGSM) [17], which leverages the gradient of the loss of input data and adjusts the input data in the direction of the gradient to create adversarial samples. Iterative variants of FGSM, such as basic iterative method (BIM) [23], projected gradient descent (PGD) [29], and momentum iterative fast gradient sign method (MI-FGSM) [10], refine perturbations to achieve stronger ASR. Carlini and Wagner (C&W) attack [4] formulates the attack as an optimization problem to generate adversarial samples with small perturbations while maintaining high attack effectiveness.

Several works have tried to promote cross-model attack transferability by adopting input transformations [40], [50] such as resizing [44], scaling [26], and their combination [39]. Different from these works, ours focuses on modeling the in-model transferability for a single perturbation to evaluate if a produced region can attack a connected region. Compared with cross-model transferable attacks, the universal adversarial perturbations (UAPs) [30] are more related to our motivation as

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UAPs focus on generating a single perturbation for all possible inputs of a model. UAP was the very first work highlighting the susceptibility of DNNs to perturbations that transcend individual instances. UAP achieves this by amalgamating perturbations learned during each iteration. The optimization approach utilized in this quest for minimal noise is borrowed from DeepFool [31], a widely used ℓ_2 adversarial attack method. Our CLT is defined on a smaller local region instead of the whole dataset, but with a stronger constraint that any point within the region should be proven vulnerable to single perturbation. All previous works measure the performance of their methods with ASRs or robust accuracy, which quantifies the proportion of images within a test dataset that the adversary can successfully deceive the target neural networks.

B. Neural Network Verification

In the realm of neural network verification, extensive efforts have been made to guarantee the robustness of deep learning models through the conceptualization of robustness as a fundamental property that can be rigorously verified. Early verification tools predominantly relied on the utilization of mixed integer linear programming (MILP) and satisfiability modulo theories (SMTs) solvers [11], [21], [36].

Nevertheless, in light of efficiency concerns, recent studies have shifted toward the adoption of convex relaxation techniques, including linear relaxation [12], [13], [41] and semidefinite relaxation [7], [33]. To further enhance the efficiency and scalability of robustness verification, recent advancements have applied bound propagation and branching algorithms (BAB) [1], [3], [28], [38], [48], along with Lagrangian decomposition methods [2], [8] and other optimizations [45].

Besides the verification technologies, randomized smoothing [5], [34] is a rather different and parallel line of research, by providing only certified guarantees constructed with high probability. Lipschitz-based network verification [24], [47] methods require the Lipschitzness property of neural networks. We focus on the typical linear programming-based neural network verification methods to provide a more persuasive metric to analyze the behavior of adversarial attacks.

We utilize neural network verification technology to accomplish our objective. Instead of verifying the robustness property of the specific neural network, we employ verifiers to generate certificates for the vulnerability of an ideal region rather than a particular point when faced with a given perturbation. In this study, the accuracy and time required for estimating the maximum vulnerable bound depend heavily on the choice of the verifier. We select GCP-CROWN [48] as our verification tool. This choice is motivated by its remarkable efficiency and extensive support for various types of specifications.

III. Problem Statement

Before delving into the details of our work, we first clarify necessary notions that will be used throughout this article. Given a L-layer neural network as a function $f\colon \mathbb{R}^{d_0} \to \mathbb{R}^{d_c}$, where d_0 is the input size and d_c is the number of classes. Consistent with previous research, we adopt the rectified linear unit (ReLU) activation function as the activation function of choice in our model.

A. Neural Network Robustness Verification

We define an input instance for DNN robustness verification as a tuple $\langle f, \boldsymbol{x}, \mathcal{C}\rangle$. Regularly, the adversarial example $\boldsymbol{x}'$ for the victim $\boldsymbol{x}$ is constrained within a region $\mathcal{C} \subset \mathbb{R}^{d_0}$. The verification target is to prove that for any point $\boldsymbol{x}' \in \mathcal{C}$, the classifier f should return the true label t. In other words, the labels' logits d_t should satisfy the following:

$$\forall y_i \neq t, d_{t,y_i}(\boldsymbol{x}') = f_t(\boldsymbol{x}') - f_i(\boldsymbol{x}') \geq 0. \tag{1}$$

This constraint forces a positive margin between the ground-truth class t and all incorrect classes $y_i \neq t$. The verifiers are designed to compute the minimal certification error $\min_{\boldsymbol{x}' \in \mathcal{C}} d_t(\boldsymbol{x}')$ in the allowed perturbation region. The property of robustness is proved if the minimum value always keeps the following:

$$\min_{\boldsymbol{x}' \in \mathcal{C}, y_i \neq t} d_{t,y_i}(\boldsymbol{x}') \geq 0 \tag{2}$$

we omit the LHS of the inequality as $d_t(\boldsymbol{x}')$ when considering the overall constraint over all incorrect predictions in the following part.

B. Vulnerable Proximal Region

As stated in the introduction, our metric measures the ability of a single perturbation, δ, to fool a model, f, not just on the original input, x, but across an entire local region, S, that surrounds it. The perturbation δ can either be produced under the constraint of a fixed length under max-norm $||\delta||_p \leq \epsilon$ [17], [29], or a minimal perturbation which is sufficient to mislead the target model [31].

Then, the vulnerable proximal region can be generally defined as a ℓ_p ball proximal region $\mathcal{S} := \{\boldsymbol{x}' \,|\, ||\boldsymbol{x}' - \boldsymbol{x}||_p \leq \epsilon\}$, the property differs from the one for robustness. For any point in this region, the perturbation can mislead the neural network to an unwanted label. Thus, we have the following:

$$\max\ [f_{i\neq t}(\boldsymbol{x}' + \delta) - f_t(\boldsymbol{x}' + \delta)] \geq 0, \forall \boldsymbol{x}' \in \mathcal{S}. \tag{3}$$

The vulnerable proximal region defined above describes the interaction between a produced perturbation and the proximal region near the victim data point. Our goal is to find the maximum possible radius, ϵ, for the region S such that the perturbation δ remains effective everywhere inside it. This radius then becomes our measure of the perturbation's strength

$$\begin{gathered} \text{maximize } \epsilon \\ \text{s.t } \max[f_{i\neq t}(\boldsymbol{x}' + \delta) - f_t(\boldsymbol{x}'+\delta)] \geq 0, \forall \boldsymbol{x}' \in \mathcal{S} \\ \mathcal{S} := \{\boldsymbol{x}' \,|\, ||\boldsymbol{x}' - \boldsymbol{x}||_p \leq \epsilon\}. \end{gathered} \tag{4}$$

Intuitively, this definition claims that for a perturbation δ generated on the victim data point $\boldsymbol{x}$, as ϵ increases, the perturbation δ is capable of affecting the classification results within a broader range. In other words, a larger ϵ implies a stronger local effect of the perturbation δ. Compared to the conventional

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metric of ASRs, which only provides a binary indication (true or false) for a specific data point, our proposed measurement advances in two key aspects. First, we output the maximum radii of vulnerable proximal regions to perturbations, offering a more diverse and precise quantification of these successfully attacked cases, moving beyond simplistic binary results. Next, we broaden the traditional perturbation analysis paradigm from a single data point to a continuous region, utilizing verification tools, allowing us to assess the robustness of the neural network across a region of inputs.

C. CLT

The vulnerable proximal regions serve as a certification for a certain perturbation to transfer from victim data points to their proximal points. We then define the *pointwise certified local transferability* (PCLT) over a single adversarial perturbation δ with the radius of the maximum vulnerable proximal ℓ_p ball of the victim point x

$$\begin{aligned} &\mathrm{PCLT}(x,\delta) := \max\ \epsilon \\ &\text{s.t. } \max[f_{i\neq t}(\boldsymbol{x}'+\delta) - f_t(\boldsymbol{x}'+\delta)] \geq 0, \forall \boldsymbol{x}' \in \mathcal{S} \\ &\mathcal{S} := \{\boldsymbol{x}' \,|\, ||\boldsymbol{x}' - \boldsymbol{x}||_p \leq \epsilon\} \end{aligned} \tag{5}$$

where the radius ϵ follows the constraint defined in (4).

From this perspective, we extend the metric from a single perturbation-radius pair to the interaction between attack methods and neural network models by averaging the impact on multiple data samples, similar to how ASRs are calculated. The *average certified local transferability* (CLT) of an attack algorithm $\mathcal{A}$ is defined as the mean of radii $\{\epsilon_i\}$ for vulnerable proximal regions $\{\mathcal{S}_i\}$ for classifier f, with computed perturbations $\{\delta_i\}$ from samples $\boldsymbol{x}_i \in D$ in the test set D by $\mathcal{A}$

$$\mathrm{CLT}(D,\mathcal{A}) := \frac{1}{n}\sum_{i=1}^{n} \mathrm{PCLT}(x_i, \mathcal{A}(x_i)). \tag{6}$$

Therefore, the *CLT* derived from the maximal vulnerable proximal regions offers a more subdivided measurement for attack methods, as compared to ASRs.

IV. Approach

In this section, we focus on the challenge of precisely and efficiently calculating the maximum vulnerable radii for perturbations to evaluate their local transferability.

A. A Reduction to DNN Verification

The constraint optimization for solving PCLT value in (5) has a similar structure but involves additional complexities. To address this, we establish a reduction from the problem defined by (5) to a neural network verification problem, specifically from the PCLT Solving problem to the Neural Network Robustness Verification problem. In essence, a reduction is a standard technique where we rigorously transform our specific problem into a well understood, general one. This allows us to solve our task by applying the powerful, existing tools already built for the general problem of neural network verification.

To demonstrate this, we convert an instance of PCLT Solving into an instance of Neural Network Robustness Verification. Given a problem instance of PCLT Solving, represented as $\langle f, \delta, \boldsymbol{x}, \mathcal{S}\rangle$, we define a new region $\mathcal{S}_\delta$ through an additive transformation

$$\mathcal{S}_\delta = \{\boldsymbol{x}' + \delta \,|\, \boldsymbol{x}' \in \mathcal{S}\} = \{\boldsymbol{x}' \,|\, ||\boldsymbol{x}' - (\boldsymbol{x}+\delta)||_p \leq \epsilon\}. \tag{7}$$

Letting $\boldsymbol{x}_\delta = \boldsymbol{x} + \delta$, we can treat $\langle f, \boldsymbol{x}_\delta, \mathcal{S}_\delta\rangle$ as an instance of Neural Network Robustness Verification in a targeted setting. This new targeted and perturbed formulation imposes a weaker constraint compared to the original Neural Network Robustness Verification problem, reflecting the shift in the region and the associated perturbation. Similarly, we select any point $\boldsymbol{x}'_\delta \in \mathcal{S}_\delta$. In the targeted version, the property we need to prove is that all samples are not classified into a specific target class label t, to satisfy the constraint

$$\max[f_{i\neq t}(\boldsymbol{x}'+\delta) - f_t(\boldsymbol{x}'+\delta)] \geq 0, \forall \boldsymbol{x}' \in \mathcal{S}. \tag{8}$$

As the maximum logit always comes from the current label c, the constraint is equivalent to

$$d_{c,t}(\boldsymbol{x}'_\delta) = f_c(\boldsymbol{x}'_\delta) - f_t(\boldsymbol{x}'_\delta) \geq 0 \tag{9}$$

always holds, or in other words, we have the following verification target:

$$\min_{\boldsymbol{x}'_\delta \in \mathcal{S}_\delta} d_{c,t}(\boldsymbol{x}'_\delta) \geq 0. \tag{10}$$

This modification can be easily achieved by changing the output layer of verifiers [38]. As a result, we proved that our constraint in our optimization problem can be regarded as a neural network verification problem, which is generally NP-complete [21]. From this perspective, we can regard $\boldsymbol{x}_\delta$ as a pseudovictim against the targeted attacks into the original ground truth label t for $\boldsymbol{x}$.

B. Oracle-Based Search With Verifiers

Some existing works raise a similar issue for searching a property-keeping space near an input sample. Previous verifiers [5], [12], [36], [41] report experimental results of maximum robust bounds as a part of the evaluation setup, but with a typical linear search or binary search to determine the minimum distortions. Some researches [25], [27] introduce a label-constrained bound propagation method to optimize nonuniform bounds against adversarial attacks. Indeed, propagation-based methods can be effective in assessing the robustness of models, but they often lack precision. The maximum bounds derived from these methods may still be far off from the optimal bound, especially when there is no prior knowledge or information on the tight bound. A verifier is also requested for external information to accelerate some search progress [9], [20]. However, these works focus on searching bounds for adversarial examples and lack the awareness of attack targets. Consequently, their ability to incorporate the latest state-of-the-art verifiers for broader DNN architecture support and improved running speed is limited.

Our task differs from previous work in two aspects. First, our certification property is different from typical ones as stated.

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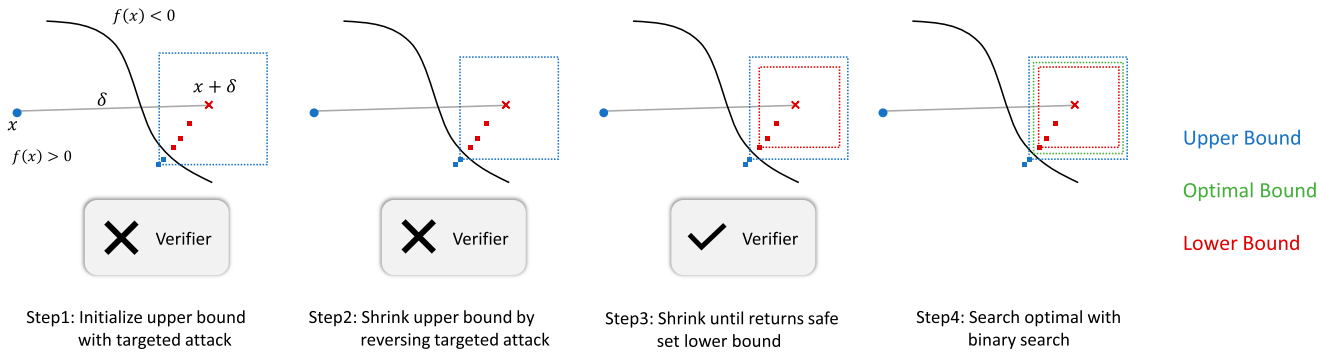


Fig. 2. Overview of RAOS via a binary classification example. The dotted line represents the notions in the corresponding color. The black curve denotes the classification boundary. The color of dots, crosses, and squares indicates their true labels, and the blue dot is the original victim data point. The gray straight line is the corresponding perturbation δ acted on $\boldsymbol{x}$. We obtain the square dots via a minimal distance-based targeted attack method $\mathcal{A}$.

Most target-specific methods are not compatible with our setting. Second, our task needs to estimate the maximal vulnerable bound as precisely as possible to quantify various attack methods. Thus, the framework should be search-based to make the most of the verifiers to obtain tight bounds.

Moreover, agnostic to the choice of the verifier to obtain better compatibility with any potential strong verifier in the future. In this way, the verifier acts as an oracle, a term for a computational tool that provides definitive "yes" or "no" answers to complex questions, without revealing its internal process., to answer whether a specification raised satisfies the property given. The oracle in our framework can be instantiated by various types of verifiers, such as complete verifiers based on MILP [36] or SMT solvers like Reluplex [21].

C. Reverse Attacks for Specifications

Verifiers are often computationally expensive. Therefore, we leverage insights from minimal distance-based attacks like C&W [4], DeepFool [31], and FAB [6]. These methods identify the smallest possible perturbations, providing excellent estimates for the maximum vulnerable bounds.

By treating $\boldsymbol{x}_\delta$ as a pseudovictim data point, we can perform targeted vision of those attacks $\mathcal{A}$, forcing the pseudovictim to return to the classifying region of ground truth class label t. The generated pseudoperturbation δ'_t indicates the direction and length under ℓ_p norm for potential minimal distortion. In this way, we can establish the first upper bound estimation of ϵ, then reversely visit the iteration steps of iterative search-based attack methods such as DeepFool. These steps are computed with a gradient descent algorithm with an adaptive step size that is automatically chosen at each iteration. Since the maximal bound is expected to be close to the length of perturbations, we request that the verifier judges the distance from $\boldsymbol{x}_\delta$ to these inner-attack steps instead of using linear search and binary search. This will help accelerate the entire search process.

V. Implementation

We now present our novel framework, RAOS, which aims to evaluate the CLT. We provide an overview of the framework, followed by a detailed technical explanation of each component and a formal definition of the operations within our framework.

A. Overview

As illustrated in Fig. 2, the kernel workflow of RAOS is to iteratively refine the upper bound and lower bound of maximally vulnerable bounds.

First, we obtain the perturbation δ from the attack method being evaluated. Next, we use a minimal distance attack to generate a sequence of intermediate steps. Finally, we use the distances from these steps as candidate radii, which are passed to a certification tool to refine the upper and lower bounds. Once the upper and lower bounds are fixed, a binary search algorithm is employed to solve the optimal bound.

B. Neural Network Verifier

Note that our property differs from the typical robustness verification. We should first slightly replace the certification objective with a targeted one as in (10). For such a verifier $\mathcal{V}$, it receives a tuple $\langle f, \mathcal{S}_\delta, t\rangle$ as input, where the specification region $\mathcal{S}_\delta$ is defined in (7), and the target label t is the ground truth label of the original victim $\boldsymbol{x}$. The verifier $\mathcal{V}$ acts as an oracle in our setting, which only returns True or False to judge if the property holds in $\mathcal{S}_\delta$

$$\mathcal{V}(f, \mathcal{S}_\delta, t) \rightarrow \{\text{True}, \text{False}\}. \tag{11}$$

We emphasize several principles of our verifier selection and framework design for computing the CLT metric.

1) *Precision*: It requires the verifier to achieve a high verification success rate for cases where the property holds. As exact verification is NP-hard [21], weaker verifiers struggle to give a verification within a limited time, which introduces a less CLT result.
2) *Scalability*: Most MILP or SMT solvers suffer from the scalability issue [38]. Those verifiers relying on MILP or SMT solvers are limited to very small problem instances. We expect the verifier to support deeper neural networks along with more complex neural network structures such as batch-normalization and dropout. Those two principles

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drive us to leverage the state-of-the-art verifier, while several related works [27], [46] directly solve the original optimization problem stated in (5).

3) *Compatibility*: As the neural network verifiers evolve rapidly, we aim to build a framework supporting any verifiers in the future. Thus, the framework should be agnostic to verifiers, which differs from previous work [9] that leverages external information from existing verifiers for searching.

In practice, we make use of the state-of-the-art neural network verifier, GCP-CROWN [48]. GCP-CROWN is based on CROWN [49] algorithm, a bound propagation method. The branch-and-bound (BAB) procedure [3] with cutting plane generation is adopted to accelerate the verification procedure. As an off-the-shelf verification tool, it provides an effective tradeoff between solving efficiency and precision, offering better scalability than traditional SMT or MILP solvers [21], [36] while supporting various complex architectures of neural networks, including residual blocks from ResNet [18]. Typical ℓ_p norm balls can be adopted to regularize the specification.

Despite the marvelous success of DNN verifiers including GCP-CROWN, the verifiers suffer from the risk of timeout for difficult instances that are close to the optimal bound. We make a tradeoff between efficiency and precision by setting the timeout limit to 90 s, while regarding the timeout status as a sort of unsafe status. Thus, the CLT we solved is less than optimal. The difference is related to the efficiency of verifier $\mathcal{V}$ and the performance of the computer for the experiment.

C. Hint Attack Method

As we mentioned in Section IV-B, we visit the steps of an attack method to get cheap and better specifications rather than a typical binary search. In this section, we introduce the specifics of the hint attack method we have selected.

The hint attack mechanism follows the intuition that the optimal bound is close to the calculated minimal distortion from a proper attack method. Formally, we define the hint attack method as a distance-minimized targeted attack

$$\mathcal{A}(\boldsymbol{x}_\delta, p, t) \rightarrow \{\boldsymbol{z}_i | 1 \leq i \leq k\} \tag{12}$$

where p is the dimension in ℓ_p, and t is simultaneously the target attack label and the label of $\boldsymbol{x}$. $\mathcal{A}$ returns an ordered set of steps from each iteration. We then calculate the distance under ℓ_p norm by $\epsilon_i = ||\boldsymbol{z}_i - \boldsymbol{x}_\delta||_p$. This distance is employed to construct the specification region S_{δ_i} for verifier $\mathcal{V}$. As $\epsilon_i < \epsilon_j$ holds for any $i < j$, the specification regions also satisfy $S_{\delta_i} \subset S_{\delta_j}$ for any $i < j$.

In practice, we extend DeepFool [31] to the ℓ_∞ norm while preserving its application on the ℓ_2 norm. Moreover, we enable targeted attacks for DeepFool to satisfy our requirements. A scaling parameter γ is used to control the converging speed by multiplying the perturbation generated each step with γ.

Algorithm 1: Solve Maximal Vulnerable Bound.

Input: f, $\boldsymbol{x}$, ground truth label t, hint attack $\mathcal{A}$, verifier $\mathcal{V}$, candidate attack $\mathcal{M}$, threshold d, strong distance-constrained attack $\mathcal{P}$

Output: Maximal Vulnerable Bound ϵ

```
Initialize lower = 0
Compute δ, x_δ with M
if argmax_j f_j(x) ≠ t or argmax_j f_j(x_δ) == t then
    return ε = 0        ▷ classification error or attack failure
Compute S_{δ_0} ⊂ S_{δ_1} ··· ⊂ S_{δ_k} with A
upper = ε_0                                   ▷ from S_{δ_0}
for i ∈ {k, k − 1, ..., 1} do
    if P successfully attack x_δ within ε_i then
        upper = ε_i
        if V(f, S_{δ_i}, t) == True then
            lower = ε_i
        else
            upper = ε_i
ε = binary_search(upper, lower, V, d)
return ε
```

D. Iterative Bound Refinement

We now introduce the whole detailed workflow, including the bound refinement process, for refining and solving the maximally vulnerable bounds for CLT in Algorithm 1.

Our method estimates the optimal bound more efficiently and precisely than a typical binary search. By using anchors from the hint attack's steps, our approach significantly reduces the number of calls to the verifier, which is the most computationally expensive part of the framework. Detailed analysis for effectiveness can be found in Table II. We also employ strong distance-constrained attack methods to prevent verifying unnecessary properties. The threshold d is set as 10^{-3} to control the precision of the bounds estimated.

VI. Experiments

In this section, we evaluate various attack methods with our proposed metric under a range of settings. We also discuss several novel findings from this new perspective of neural network robustness.

A. Experiment Setup

We develop the project in Python. We implement most attack methods with torchattack [22], a library offering a handful of adversarial attacks. Experiments ran on an AMD Ryzen 9 5950X processor with 16 cores and 128 GB RAM. All GPU-related computations are finished with an NVIDIA RTX 3090 GPU with 24 GB memory.

B. Datasets and Target Models

Following existing works on neural network verification [7], [38], we conduct experiments on MNIST dataset and CIFAR10

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TABLE I
OVERVIEW OF EXPERIMENTAL RESULTS

Dataset	Model	#Para.	Adv. Training	%Acc.	CLT × 10^3/%Robust Acc.					
					FGSM	PGD	MI-FGSM	C&W	FAB	UAP
MNIST	MLP-9-100	160k	-	99	6.18/13	14.60/8	11.4/10	<0.01/2	0.11/3	-
			0.1	99	2.24/89	2.62/85	2.55/86	<0.01/7	<0.01/86	-
	CNN-Small	166k	-	100	4.77/65	11.74/15	10.82/24	<0.01/3	<0.01/16	-
			0.1	99	0.38/96	0.60/95	0.75/95	<0.01/7	<0.01/95	-
CIFAR10	ResNet-2b	215K	-	81	0.42/24	0.63/18	0.68/17	<0.01/0	<0.01/16	0.89/28
			2/255	77	0.37/63	0.39/63	0.38/63	<0.01/0	<0.01/60	0.71/62
	CNN-A	112k	-	83	0.99/27	1.20/17	1.27/17	<0.01/0	<0.01/19	1.82/31
			2/255	70	0.42/55	0.43/54	0.43/54	<0.01/68	<0.01/48	0.32/67

Note: The Certified Local Transferability (CLT) is listed along with robust accuracy for comparison. The under for Adv. Training indicates the model is adversarially trained with perturbations of that length.

dataset. For the former, our evaluation is based on two architectures: *MLP-9-100* and *CNN-Small* from [41], [43] with 160k and 166k parameters. Each architecture has two sets of parameters: the vanilla-trained one and the adversarial-trained one [43]. For the latter, we also investigate two architectures: *CIFAR-CNN-A* and *ResNet-2b* from Wang et al. [38] with 215k and 112k parameters. Vanilla- and adversarial-trained parameters are also provided for this task. The large size and various architectures of the models evaluated are sufficient to demonstrate the scalability of our method. Considering the computational cost of neural network verification, following most previous work on neural network verification [35], [38], [41], [43], a feasible practice focuses on evaluating the first 100 examples from both datasets instead of the whole test set.

C. Attack Methods

As a novel metric to evaluate adversarial attacks, we mainly consider typical white-box attack methods such as PGD, FGSM, MI-FGSM, C&W, and fast adaptive boundary (FAB) attacks, categorized into distance-minimized and distance-constrained. Specifically, we also employ UAPs, to investigate the difference between global transferability and local transferability. The default distance constraint for attack methods to be evaluated is 0.1 for the MNIST dataset and 2/255 for the CIFAR10 dataset. For UAPs, we set the maximal length of perturbation to 16/255, to discover their performance on CLT metric with close results on robust accuracy metric to FGSM, MI-FGSM, and PGD. We present an overall result of CLT in Table I. For better comparison, we compute our metric under ℓ_∞ norm for all ℓ_p-based attack methods where $p = 1, 2, \infty$ may hold. We should note that the framework we apply supports measurement under other ℓ_p norms.

D. Empirical Analysis and Insights on Attack Transferability

1) Overview: As illustrated in Table I, our proposed metric, CLT, exhibits a notable correlation with the robust accuracy metric, as the CLT value for misclassified samples and robust samples is zero. However, the true value of CLT lies in its ability to reveal critical distinctions where robust accuracy falls short. For instance, on the vanilla-trained MLP-9-100 model, PGD's CLT of 14.60 is more than double that of FGSM's 6.18, even though both attacks are highly successful at deceiving the model. This powerfully demonstrates that iterative optimization finds a perturbation with substantially stronger local transferability, a nuance that simple ASRs cannot capture. Our analysis of MI-FGSM further solidifies this point. As a variant of PGD that incorporates momentum to stabilize the iterative update directions, MI-FGSM shows a CLT performance that is highly comparable to, and in some cases slightly exceeds, that of PGD across our experiments (see Table I). This suggests that the momentum technique, well known for helping attacks escape shallow local optima, also contributes to generating perturbations with powerful local transferability. This phenomenon also demonstrates the dependency of CLT on robust accuracy and similar metrics.

2) Distance-Minimized Attacks and Distance-Constrained Attacks: We note that the CLT of the C&W attack is poor even with a robust accuracy close to 0. We consider this case as a benign one since the C&W attack is a distance-minimized attack that aims to find minimal distortion to deceive the classifier. This highlights a fundamental dichotomy between two primary attack strategies, which we illustrate conceptually in Fig. 3.

For the distance-constrained FGSM attack, the process begins by calculating the gradient of the model's loss with respect to the input x. It then takes the sign of this gradient and scales it by a fixed perturbation budget, ϵ, to create δ_{FGSM}. As depicted in Fig. 3, this method's objective is not to find the closest boundary point, but to maximize the loss within its allowed budget. This often results in a large perturbation that pushes the input far away from the decision boundary, which in turn ensures the same perturbation is effective for a large surrounding region, resulting in a high CLT.

In contrast, distance-minimized attacks like C&W and FAB are designed to find the shortest possible escape vector, landing the adversarial example just across the boundary. Our results for the FAB attack in Table I provide powerful corroborating evidence, as it also consistently yields near-zero CLT values. Consequently, this highly optimized, minimal perturbation has a very small transferable region and therefore a low CLT. The fact that two distinct, state-of-the-art distance-minimized attacks both register extremely low CLT scores illustrates the

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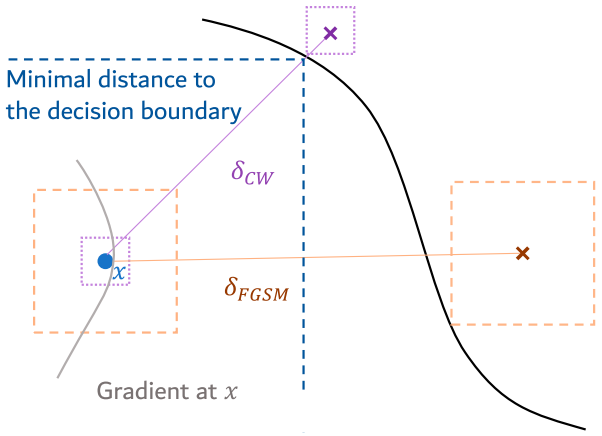


Fig. 3. Conceptual comparison of distance-constrained attacks like FGSM and distance-minimized attacks like C&W. Both attacks start from the same point x. The FGSM attack uses its perturbation budget to push the sample far across the decision boundary, while the C&W attack finds the minimal perturbation to just cross it. This objective difference leads to a large CLT for FGSM and a near-zero CLT for C&W.

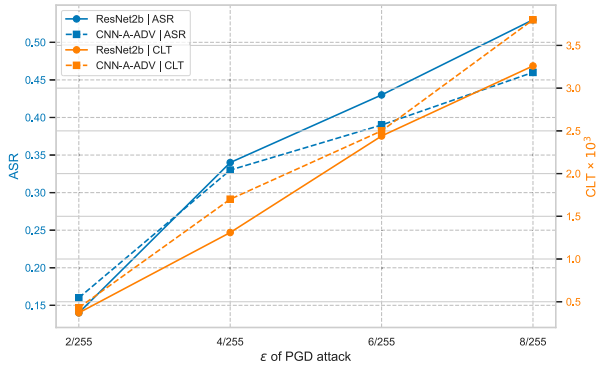


Fig. 4. Trends of ASR and CLT varying with the magnitude of perturbations generated from PGD. The victim model is adversarially trained with a perturbation of 2/255.

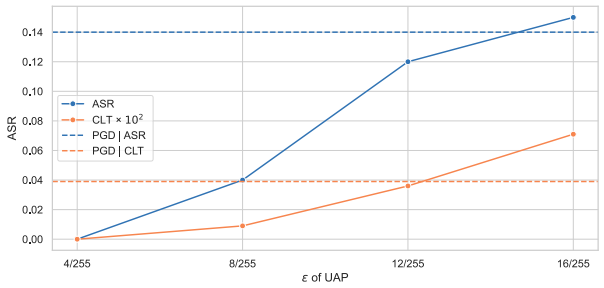


Fig. 5. Trends of ASR and CLT varying with the magnitude of perturbations generated from UAP. The dotted lines are referral values from PGD attack with $\epsilon = 2/255$.

dual explanatory power of our CLT metric: a high CLT indicates a potent distance-constrained attack, while a low CLT can signify a successful and precise distance-minimized attack.

3) UAP Works Locally: We are curious about UAPs, which are designed to fool most samples in the dataset at the same time with a single perturbation. The motivation of UAPs is close to ours in considering a single perturbation exerting multiple samples. The main divergence is that our metric targets a continuous local space, while UAPs aim for a discrete dataset. Intuitively, we expect UAPs to outperform most attack methods for their related objective to ours. As shown in Table I, UAPs perform much better than FGSM and PGD in most cases, even achieving the highest robust accuracy, meaning the lowest ASR. We believe this is a significant finding: UAPs, designed for global, cross-sample transferability, also exhibit remarkably strong local transferability in continuous space. This suggests a link between the mechanisms for global and local transferability and provides a new path for improving future UAP generation methods, for instance, by incorporating the maximization of CLT as a direct optimization objective.

4) Larger Perturbation Provides Better Local Transferability: We conduct an experiment to investigate the relationship between CLT and the maximal distance allowed for distance-constrained attacks. We enhance the perturbation produced by PGD to attack *ResNet2b* and *CIFAR-CNN-A*, two target models adversarially trained with PGD perturbation of 2/255. Fig. 4 depicts how the ASRs and CLT change with varying the maximum tolerance of perturbation in PGD attack across 2/255, 4/255, 6/255, and 8/255. Both metrics for the two models manifest a similar trend, where the ResNet-2b and CIFAR-CNN-A models exhibit decreasing accuracy along with increasing local transferability as ϵ escalates. We can infer from the trend that attack algorithms tend to compose strong perturbations against local regions if given sufficient space tolerance to search. We also provide a similar finding of UAPs in Fig. 5. This figure supports our conclusion that UAPs obtain better CLT results than attack methods such as PGD when acquiring close ASR results. The growth trend of CLT value is also greater than the ASR.

E. Efficiency and Scalability Analysis

1) Optimization via Reverse-Attack Hint: A key aspect of our framework is the efficiency of finding the maximal vulnerable bound. To this end, we introduced a reverse-attack hint mechanism to accelerate the search process, a significant improvement over the widely used binary search algorithm found in many verification studies [5], [12], [36], [41]. We offer a case study for both algorithms in Table II, where the comparison is conducted on the CIFAR-CNN-A model and a PGD attack with $\epsilon = 2/255$. The results clearly show that our proposed reverse-attack hint offers better efficiency for refining the maximum valuable bound of local transferability. It reduces the number of calls to the verifier from 51 to 21, leading to a substantial decrease in total runtime from 631 to 487 s. This demonstrates the effectiveness of our approach in creating cheaper and more precise estimations of the optimal bound by reducing the reliance on calling the computationally expensive verifier.

2) Configurable Tradeoff Between Precision and Efficiency: Despite this optimization, we acknowledge that the overall computational cost for calculating the CLT metric remains high. This is an inherent challenge, as our method's precision relies on state-of-the-art neural network verifiers, and the underlying verification problem is generally NP-Complete. However, this

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TABLE II
COMPARISON OF BINARY SEARCH AND REVERSE-ATTACK HINT

Search Method	Time (s)	# Veri. Calls
Binary Search	631	51
Reverse-Attack Hint	487	21

high cost should be viewed not as a rigid barrier but as a manageable tradeoff between precision and efficiency, which is controllable through several parameters.

One key parameter is the verifier's timeout limit. In our own experiments, we made a practical choice by setting this to 90 s. This ensures that we can obtain certified bounds within a reasonable timeframe, while accepting that the resulting CLT value might be a conservative under-estimation of the true optimal bound for difficult instances.

Another configurable parameter is the precision threshold of the final binary search stage. If we relax this precision with a larger threshold, the search will terminate in fewer iterations, thus accelerating the process. The tradeoff is that the resulting certified region will be slightly smaller or more conservative, as the search halts before maximizing the certifiable lower bound to its fullest extent. In practice, we set the precision to $1e{-}3$. These examples highlight the flexibility of our framework: users can configure it for rapid, approximate analysis or for maximum precision, depending on their specific application scenario.

VII. CONCLUSION

In this article, we introduce the concept of CLT for adversarial attacks, defined as the maximal local region near the victim points that the same perturbation can successfully attack. To solve the CLT precisely and efficiently with high usability and scalability, we regard neural network verifiers as oracles and combine them with a reverse-attack search mechanism to iteratively refine the maximally vulnerable bounds. This novel measurement provides a new perspective for investigating adversarial attacks and neural network robustness.

A. Limitations of This Work

Our framework relies on state-of-the-art verifiers. As neural network verification problem is generally NP-Complete [21], verifiers are not efficient enough to provide precise certified bound due to limited time and computing resources. Thus, the precision of our metric may be affected. This work also shares similar limitations such as time cost and dataset size with existing neural network verification techniques.

B. Broader Applications and Future Directions

Beyond providing a more nuanced evaluation of existing adversarial attacks, our CLT metric opens up several promising avenues for future research and practical applications:

1) *High-quality data augmentation for adversarial training*: This is perhaps one of the most direct applications of our work. Traditional adversarial training often augments data with single, pointwise adversarial examples. The certified vulnerable region $\mathcal{S}$ that we compute provides a rich source of diverse inputs that are guaranteed to be adversarial when perturbed by the same δ. A defender can sample multiple points x' from within this certified region $\mathcal{S}$ to create a set of high-quality training samples $\{x' + \delta\}$. Training on such a guaranteed set of varied adversarial examples, rather than a single point, could lead to models with smoother decision boundaries and significantly improved robustness against attacks that impact broader input regions.
2) *Advanced model selection*: CLT provides a fine-grained metric for model selection. When standard metrics are similar, the model with a consistently lower average CLT against strong attacks can be chosen for its superior regional robustness, which is critical for safety-sensitive applications.
3) *Novel defense evaluation and design*: The metric offers a new way to evaluate defenses. An effective defense should not only block attacks but also significantly reduce the CLT of successful ones. Furthermore, minimizing CLT can be a novel objective for designing next-generation defenses.
4) *Guidance for potent attack design*: Conversely, CLT can guide the design of more potent attacks. Attackers can aim to maximize the CLT of a perturbation, creating attacks that are more resilient to minor input variations and thus more practical in physical-world settings.

C. Potential Societal Impact

Our work provides a new metric to evaluate adversarial perturbations, which is an important issue in trustworthy AI. This metric can be guidance for developing new attacking technologies, but with defending technologies such as adversarial training, similar malicious real-world attacks can be better modeled and prevented.

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